POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2016
Programme: BE | Full Marks: 100
Course: Digital Signal Analysis and Processing | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Differentiate between signal analysis and signal processing. What are the advantages of digital signal processing over analog signal processing? Explain. 8
   b) Show that the impulse response of a BIBO stable LTI system is absolutely summable. 7
2. a) Verify the commutative property of the convolution between two signals $x[n] = a^n u[n]$, $0<a<1$ and $h[n] = u[n]$. 8
   b) Find the Z-transform of $x[n] = a^n \cos w_0 n\, u[n]$ 7
3. a) Find 8-point DFT of the sequence {1,2,3,4,5,4,3,2} using radix-2 decimation in time algorithm. 8
   b) Explain the "Multiplication of two DFT's and circular convolution" property of DFT. 7
4. a) Draw the lattice ladder diagram for the following system: 8
   $H(z) = \frac{1-0.8z^{-1}-0.15z^{-2}}{1+0.2z^{-1}+0.8z^{-2}}$, Is the system stable? 7
   b) Determine the direct form realisations of the following differential equation:
   $2y[n] = 4y[n-1] - 4y[n-2] + 6y[n-4] + x[n-1] - x[n-3]$
5. a) What is Linear phase FIR filter? Explain FIR filter design by Frequency-Sampling method. 8
   b) Design a linear FIR filter using Kiser window to meet the following specifications: 7
   $$0.99 \le |H(e^{j\omega})| \le 1.01, for\ 0 \le |\omega| \le 0.19\pi$$
   $$\le |H(e^{j\omega})| \le 0.01, for\ 0.21\pi \le |\omega| \le \pi$$
6. a) Design a digital Butterworth lowpass filter using Bilinear 8

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Transformation method to meet the following specifications:

Passband attenuation≤1.28 dB Passband edge frequency=200 Hz

Stopband attenuation≥13 db Stopband edge frequency=350 Hz

Sampling frequency=1 KHz.

b) Compare between FIR and IIR filter.

7. Write short notes on: **(Any two)**

a) Remez Exchange Algorithm

b) Z-transform and ROC

c) Casual and Non-casual system

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